

30

Time complexity

CSK/DA

$\Rightarrow$

gate

Lec-11

$\frac{0-10}{11}$
Base

~~$\frac{0-10}{11}$~~
 40%

$\frac{4-6}{11}$

$\frac{0-3}{11}$

Big O, Ω , Θ

Lec 11

Introduction to Asymptotic notations

Intro to : Big O , small o , theta ,
Omega , small omega

$\frac{0}{1}, \Omega, \Theta$

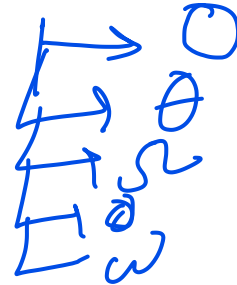
$\frac{0}{1}, \Omega, \omega, \Theta$

Ex: 1

Time complexity

2.6ms

Asymptotic notations



Big O

program or Algorithm

Ex: find the an element x in an array and return it's position?

If not present return -1

int arr[6] = { 23, 6, 27, 60, 15, 19 }
 0 1 2 3 4 5

$x = 60$

case: 1

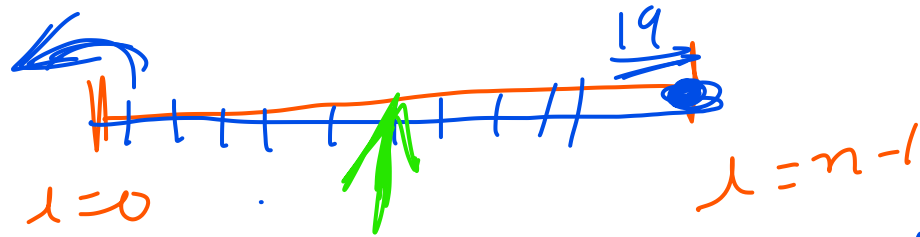
4 times

```
for (int i = 0; i < n; i++)  
{  
    if (arr[i] == x)  
        return i;  
}  
return -1
```

4
i = 0
i = 1
i = 2
i = 3

○ Case 2 $n=19$ - $l=0, l=1, l=2, l=3, l=4, l=5$
 5 times
Case 3 $n=23$ 11 times
 $l=0$; $arr[0] = 23$ $(l=0)$

wrong



○ $\Rightarrow$ max^m time
 □ $\Rightarrow$ min^m time

Case 1

worst case $\Rightarrow$ max^m time

$O(n)$

$O(n-1)$
 $O(n)$

Case 3

Best case $\Rightarrow$ min^m time

$\Omega(1)$

case 1:

Avg case $\Rightarrow$ Avg time
 $\Theta(n/2) \Rightarrow \Theta(n)$

$\Rightarrow$

$O \Rightarrow$

max^m case

$\Omega \Rightarrow$

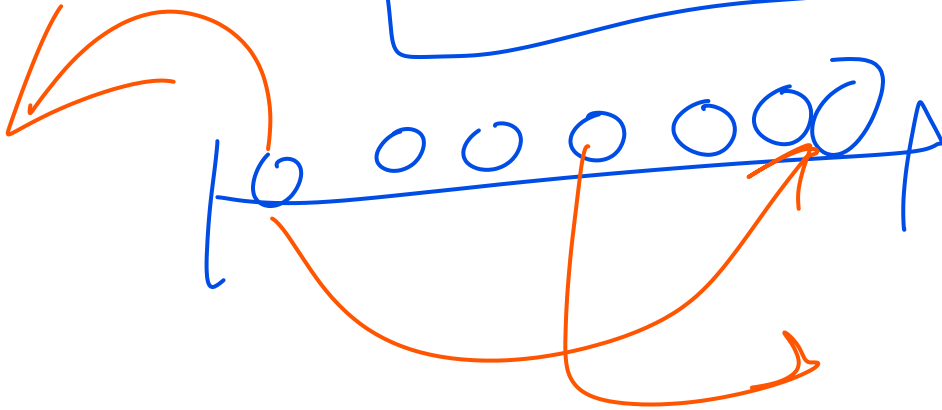
min case

Θ

$\Rightarrow$

Avg case

*



$$\underline{O, \Omega, \Theta}$$

$$\left. \begin{array}{l} \text{Time} = n \log n \\ TC = n^2 \\ TC = \underline{n \log \log n} \end{array} \right\} \begin{array}{l} \text{fn} \\ \text{in} \\ \text{the} \\ \text{form} \\ \text{of} \\ n \end{array}$$

1. Big Oh Notation (O):

Let two non negative functions $f(n)$ and $g(n)$:

Ex $\underline{TC} = O(n)$ means $\max(TC) = n$

$$f(n) = O(g(n))$$

$\max^n$ value of $f(n)$ is $\underline{g(n)}$

means $\boxed{f(n) \leq g(n)}$

Definition = $\boxed{\begin{array}{l} \text{if } f(n) \leq g(n) \\ \text{then } f(n) = O(g(n)) \end{array}}$

$$\begin{array}{l} f(n) = n^2 \\ f(n) = \log n \\ \underline{g(n) = n^3} \end{array}$$

Now more:

0 $\Rightarrow$

if $f(n) \leq c \cdot g(n)$
then, $f(n) = O(g(n))$

$c = \text{constant}$,

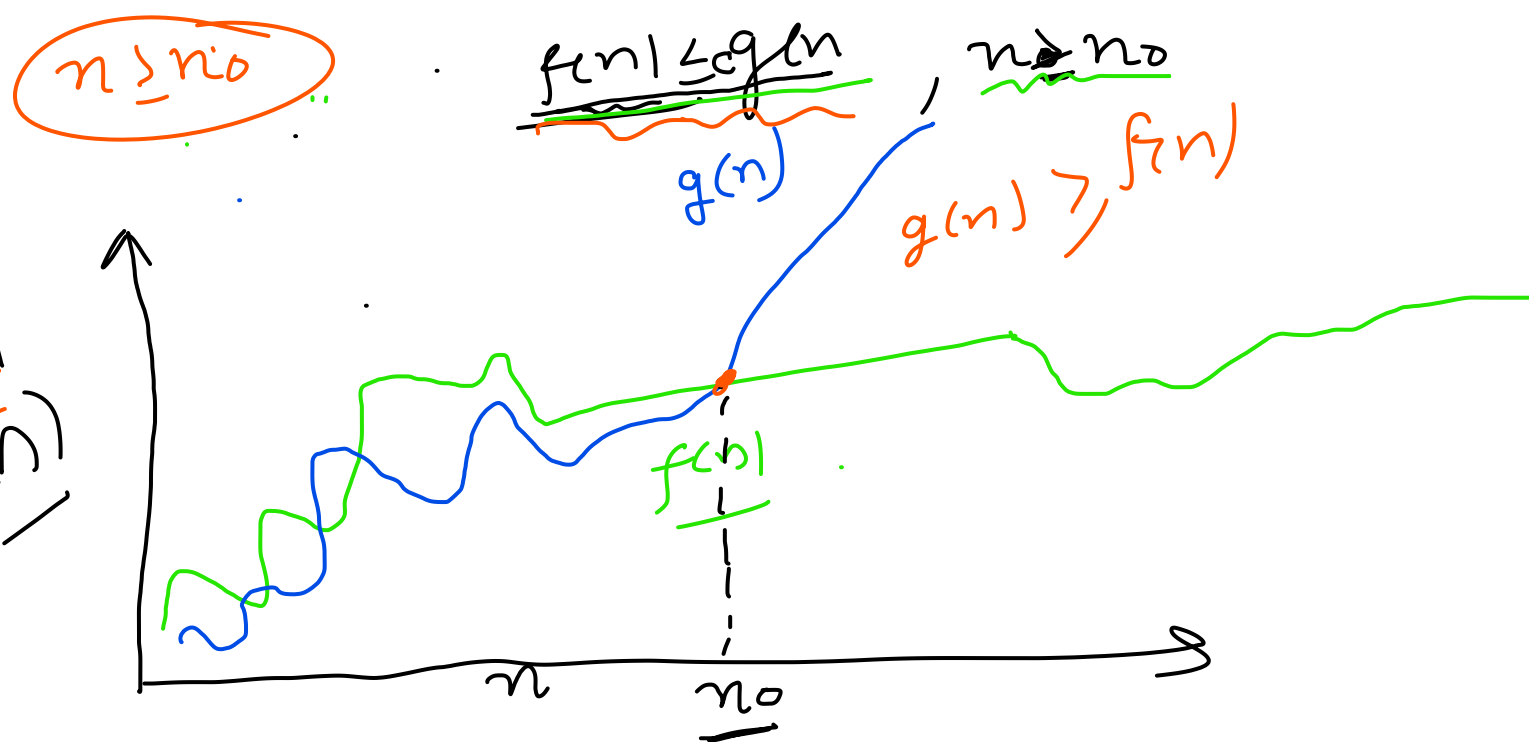
$n \geq 1$

$n \geq n_0$

$c > 0$

Graph:

$\frac{f(n)}{g(n)}$
or
 $\frac{g(n)}{f(n)}$



graph $\Rightarrow$ Big(O)

$$f(n) = n + 10$$

$$g(n) = n$$

Big O an ahi hai

$$f(n) \leq c \cdot g(n)$$

Ex:1

$$F(n) = n + 10, g(n) = n$$

$$f(n) \leq c \cdot g(n)$$

$$n + 10 \leq c \cdot n$$

$$n + 10 \leq 2n$$

$$f(n) = n + 10$$

$$g(n) = 2n$$

$$c > 0$$

$$f(1)$$

$$g(1)$$

$$11$$

$$2$$

$$12$$

$$4$$

$$13$$

$$6$$

$$14$$

$$8$$

$$15$$

$$10$$

$$16$$

$$12$$

$$20$$

$$20$$

$$21$$

$$22$$

$$22$$

$$24$$

$$f(n) > g(n)$$

$$n = 11$$

$$f(n) \leq g(n)$$

$$c = 2, n_0 = 20$$

$$n + 10 \leq c \cdot n$$

$$n + 10 \Rightarrow O(n)$$

$$n + 10$$

$$2n$$

$$O(n)$$

Ex:2

$F(n) = \underline{n^2 + n + 10}$, $g(n) = n^2$

$f(n) \leq \underline{C \cdot g(n)}$

	$f(n)$	$g(n)$
$n=1$	<u>12</u>	10

$n=2$	<u>16</u>	<u>40</u>
-------	-----------	-----------

$\underline{n^2 + n + 10} \leq \underline{C n^2}$

$f(n) = n^2 + n + 10$

$g(n) = 10n^2$

$C = 10$

$n_0 = 1$

$f(n) \leq C \cdot \underline{g(n)} \Rightarrow \underline{O(n^2)}$

$\Rightarrow f(n) = O(g(n))$

$\underline{n^2 + n + 10} = \underline{O(n^2)}$

Ex:3

$F(n)=n-10, g(n)=n+10$

$$f(n) \leq c \cdot g(n)$$

$$\underline{n-10} \leq \frac{(n+10)}{\underline{106}}$$

$$\Downarrow \\ \underline{\underline{O(n)}}$$

$$\underline{c > 0}$$

$$\begin{array}{l} n-10 \\ n+10 \\ n+20 \\ n+30 \end{array}$$

$$\begin{array}{l} n/20 \\ \underline{n/2} \end{array}$$

Ex:4

$F(n)=n+1, g(n)=n/2$

$$f(n) \leq c \cdot g(n)$$
$$f(n) = O(g(n))$$

$$n+1 \leq \frac{4n}{2} \quad c=4$$

$$f(n) \leq 4 \cdot g(n)$$
$$f(n) = O(n/2)$$
$$= \underline{O(n)}$$

$$\text{if } f(n) \leq c \cdot g(n)$$
$$f(n) = O(g(n))$$

Lecture $\Rightarrow$ Summary

O $\Rightarrow$ max
 Ω $\Rightarrow$ min
 Θ $\Rightarrow$ avg.